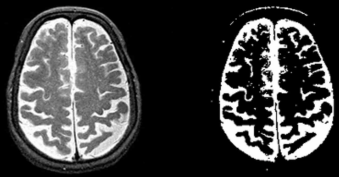
CLASSICAL COMPUTER VISION

Traditional Computer Vision

Core Techniques

- Thresholding
  Pixels $\geq$ a value $\rightarrow$ set to maximum
- Edge Detection
  Detect changes in brightness
- Segmentation
  Group regions enclosed by edges
- Curve Detection
  Link adjacent edges into shapes
- Optical Flow
  Estimate motion between frames



10.4018/978-1-5225-0140-4.ch004

Classical pipeline: Pixels $\rightarrow$ Edges $\rightarrow$ Shapes $\rightarrow$ Motion

Feature Engineering: Sobel Filters

Edge: a location where image intensity changes abruptly.

Goal: find pixels where the gradient magnitude is large.

Intuition

- Flat region $\rightarrow$ small intensity change
- Object boundary $\rightarrow$ sharp intensity change

Practical Significance

- Extract object boundaries
- Reduce data complexity

Numerical Method

$\Rightarrow$ **Sobel filter**

Treat the image as a function $f(x, y)$:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

Edge strength (gradient magnitude):

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

Sobel Filter: Vertical Edge Detection

Image Patch (I)

10	10	200
10	10	200
10	10	200

Dark $\rightarrow$ Bright

$*$

Sobel Kernel (G_x)

-1	0	1
-2	0	2
-1	0	1

$=$

Response

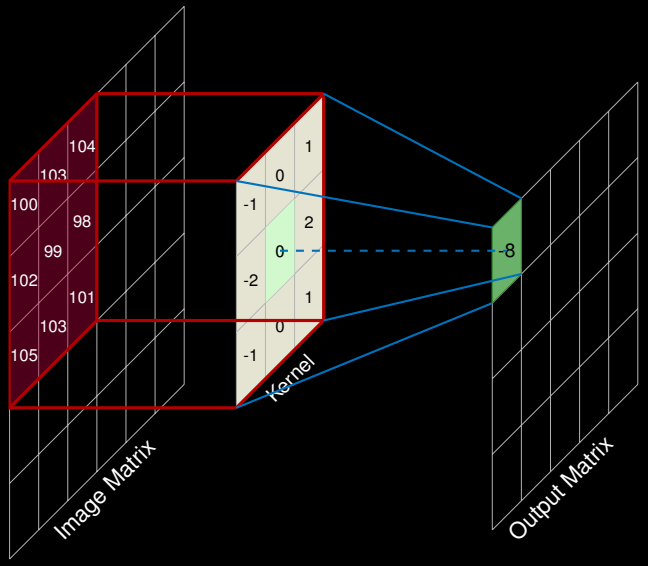
760

$$\text{Sum} = (-1 \cdot 10) + (0 \cdot 10) + (1 \cdot 200) + \cdots = \mathbf{760}$$

*The **negative weights** suppress dark pixels, while **positive weights** amplify bright pixels. A high response (760) indicates a strong **vertical edge**.*

Sobel Filter with Convolution

The convolution happens between the input image and the given kernel. It is the **sliding dot product** between the kernel and the localised section of the input image.



Feature Engineering: Template Matching

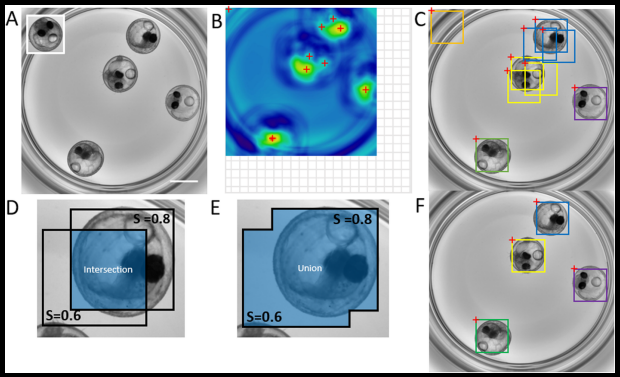
Slide the template over the image and score the match at every position. The **correlation map** on the left is that score; its brightest point is the detection on the right.



OpenCV, *Template Matching* tutorial

Feature Engineering: Template Matching

Find the embryos in the microscopy image.



Thomas & Gehrig, *Multi-Template Matching: a versatile tool for object-localization in microscopy images*, BMC Bioinformatics

TEMPLATE MATCHING - CROSS CORRELATION

